The Invariance Principle — problem 1.57

A. Engel — Problem-Solving Strategies · 1998

Starting with a point $S(a, b)$ of the plane with $0 < a < b$, we generate a sequence (x_n, y_n) of points according to the rule

$$x_0 = a, \quad y_0 = b, \quad x_{n+1} = \sqrt{x_n y_{n+1}}, \quad y_{n+1} = \sqrt{x_n y_n}.$$

Prove that there is a limiting point with $x = y$. Find this limit.